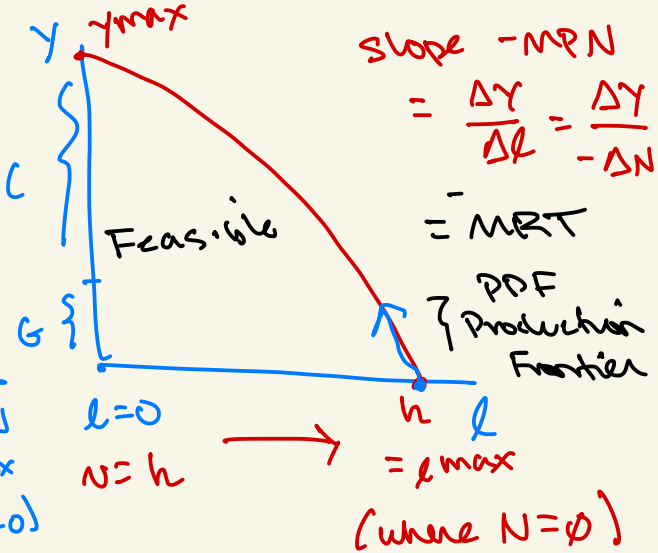
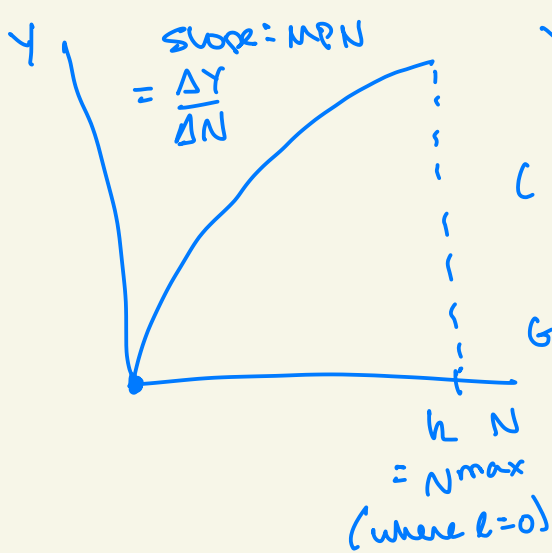
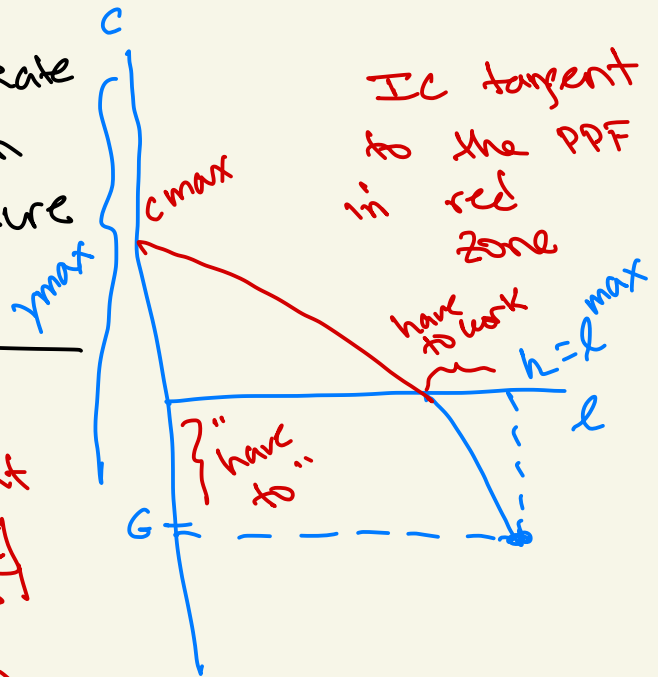


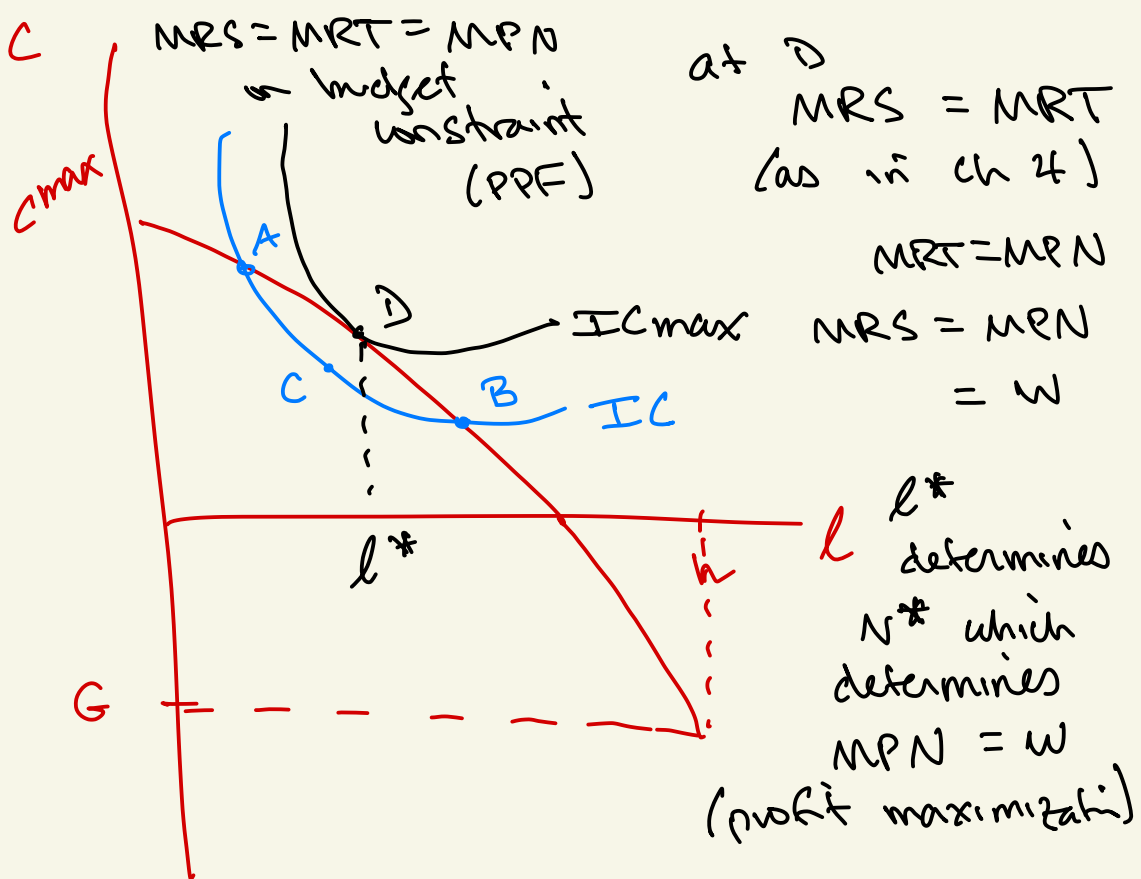



MRT: Marginal Rate of Transformation of C into leisure (Y)

$$Y = C + G$$

G government spending (exogenous)
not chosen by market



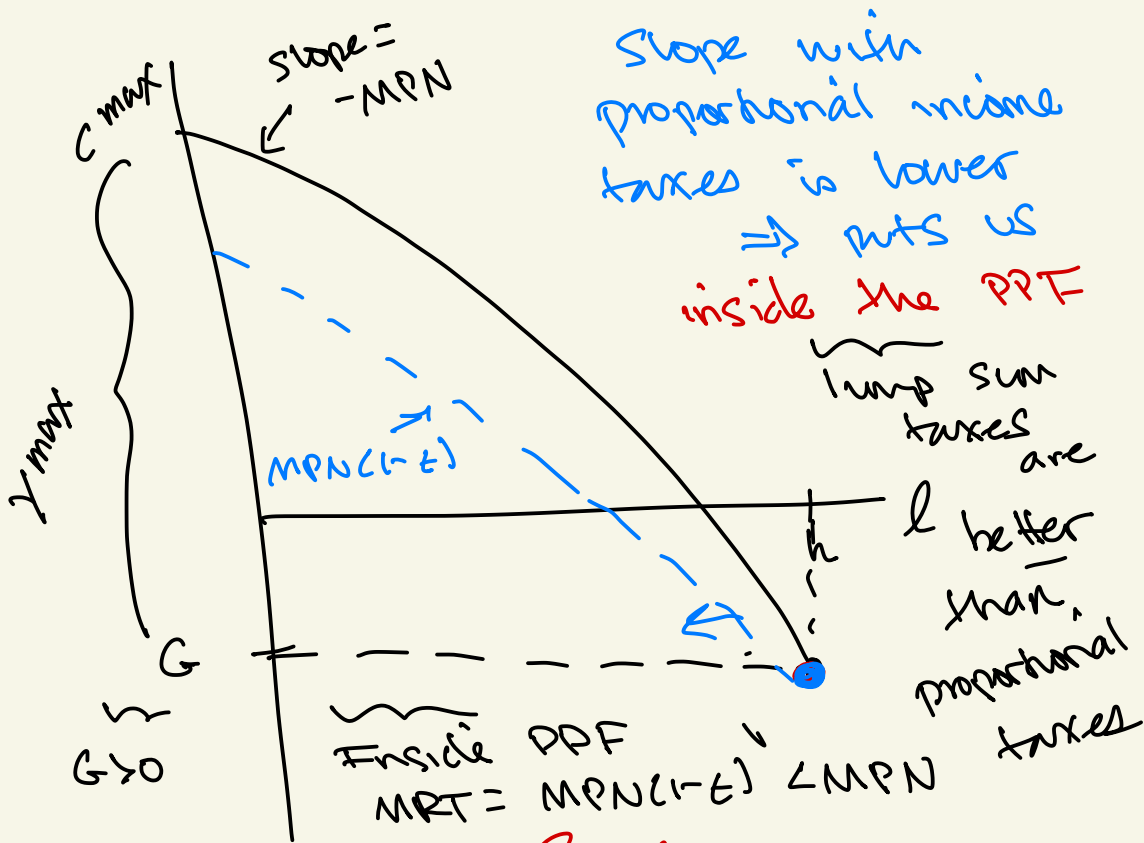


$$y_{\max} = C^{\max} + G \quad \text{where } l = 0$$

$$N = h$$

slope = $-MRT$ } rate at which leisure is turned into C

In eq'm = $-MRS$ } rate at which want to turn leisure into C



2 ways pay for G :

1. Lump-sum tax (no impact on MPN or MRT) $\Rightarrow \frac{\Delta Y}{\Delta l}$ unaffected by ΔT

2. Proportion taxes t . $Y = \text{Revenue}$

$$\frac{\Delta C}{\Delta l} \downarrow \text{ as } t \uparrow$$

work $\uparrow$ but government taxes a % so $C \uparrow$ less

* Proportional taxes are less efficient than lump-sum taxes because they distort / reduce $\frac{\Delta C}{\Delta L}$ } slope of PPF in the range of household choices

* anything that makes anyone worse off without making someone else better off is "inefficient"

$$\underbrace{Y^{\max}}_{2K^{\alpha}h^{\beta}} = C + G$$

$\downarrow$ $\uparrow$

Ch 5 : 3 subjects

1. Solve for w

2. Define efficiency as a
Pareto Optimum

Pareto's Definition:

Can't make anyone better
off without making someone
else worse off.

e.g. Proportional taxation that earns
the same amount as lump
sum taxes

$\Rightarrow$ reduces $C^{\max}$
without making anyone
better off (inefficient)

* market uses " t " but we
can measure market inefficiency
by comparing it to "T" case

3. See if the model
can explain business cycle
data (ch 3)

Example : Preferences shift

toward C-goods
eg $U = a \ln C + b \ln L$

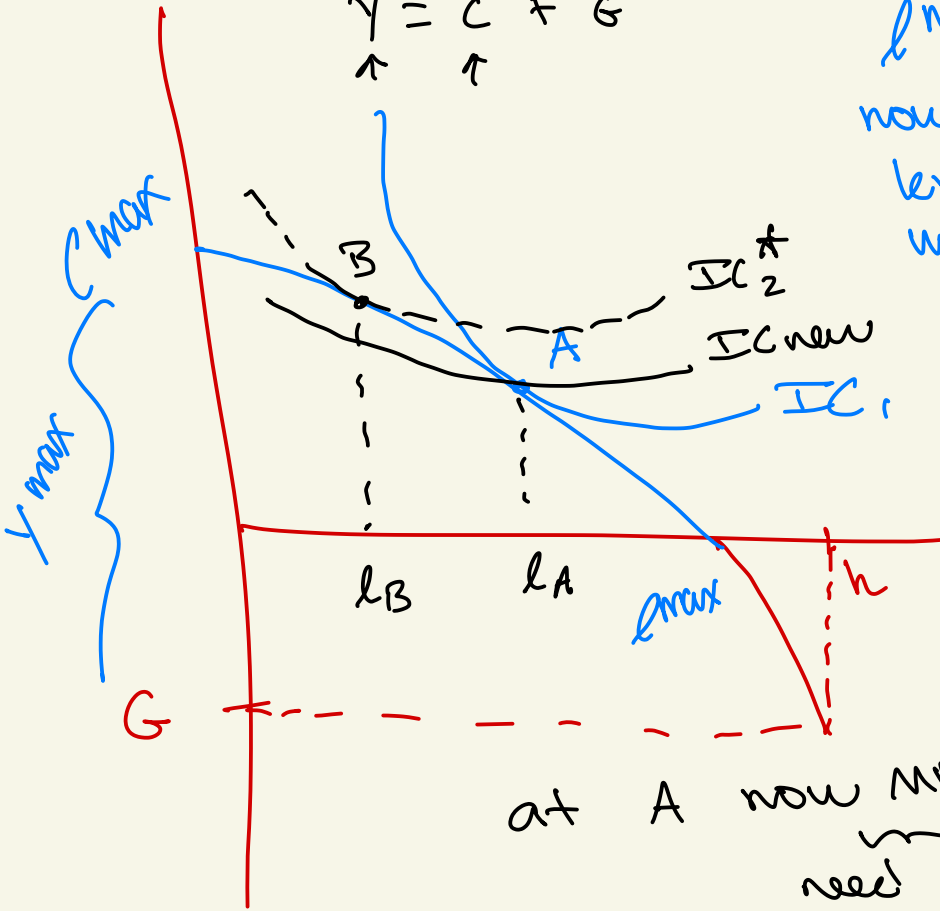
$$\downarrow MRS = \frac{MU_C(L)}{\uparrow MU_L(L)}$$

IC flatter

$$Y = C + \bar{G}$$

$a \uparrow$

$\downarrow$ max is
now max
leisure
we can
choose



$\Delta U?$
 $L \downarrow C \uparrow$
(NPF)
 $\sim$
 $Y \uparrow$

at A now $MRS < MRT$

$N \uparrow MPN \downarrow$
 $\therefore w \downarrow$

$\frac{MU_C(L) \uparrow}{MU_L(L) \downarrow}$ need
 $\left. \begin{array}{l} L \downarrow \\ C \uparrow \end{array} \right\}$

* Note in this example
I implicitly assumed lump-sum
taxes (T)

$\Rightarrow$ assumed to be on
the PDF (not inside
it)